Search for Σ^0 Hyperon Production in Anti-Neutrino Interactions at MicroBooNE

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1 Introduction

Strange-particle production in neutrino–nucleus interactions provides a sensitive test of the electroweak current structure and nuclear effects at energies relevant to the oscillation experiments of the short- and long-baseline programmes. The neutral Sigma hyperon, Σ^0, is the lightest strange baryon accessible via charged-current (CC) anti-neutrino scattering off nucleons and has not yet been observed in neutrino experiments.

The Σ^0 carries isospin $I = 1$, strangeness $S = -1$, and decays electromagnetically with a branching fraction of essentially 100%:

$$\Sigma^0 \;\rightarrow\; \Lambda + \gamma \qquad (\tau \approx 7.4 \times 10^{-20}\ \mathrm{s})\,. \tag{1}$$

The subsequent weak decay $\Lambda \rightarrow p + \pi^-$ (branching fraction 63.9 %) yields a fully charged-particle final state alongside the hard photon, making the reaction observable in a fine-grained detector. The complete signal topology studied here is therefore

$$\bar{\nu}_\mu + N \;\rightarrow\; \mu^+ + \Sigma^0\,, \quad \Sigma^0 \rightarrow \Lambda + \gamma\,, \quad \Lambda \rightarrow p + \pi^-\,, \tag{2}$$

giving four visible final-state objects: a muon (μ^+), a proton (p), a charged pion (π^-), and a photon (γ).

The associated production mechanism $\bar{\nu}_\mu\, p \rightarrow \mu^+\, \Sigma^0$ proceeds via the charged weak current with strangeness change $|\Delta S| = 1$ and is Cabibbo suppressed relative to quasi-elastic scattering by a factor $\tan^2\theta_C \approx 0.05$. Theoretical predictions for the total cross section span a range of $\mathcal{O}(10^{-40})$ cm^2 at neutrino energies of ~ 1 GeV. A positive observation, or a stringent upper limit, constrains hyperon-production models that feed into flux-averaged oscillation analyses.

Additionally, the Σ^0 search requires simultaneous reconstruction of three tracks and one electromagnetic shower, exercising the full particle-identification capabilities of a liquid-argon time projection chamber (LArTPC) in a way that benefits reconstruction algorithm development.

1.1 Cross-Section Theory and Intranuclear Rescattering

The elementary cross section for $\bar{\nu}_\mu\, p \rightarrow \mu^+\, \Sigma^0$ is derived from the $V\!-\!A$ structure of the electroweak Standard Model. The hadronic matrix element is parametrised by six Lorentz-invariant form factors consistent with SU(3) flavour symmetry [8, 10]:

- Two isovector *vector* form factors $f_1(Q^2)$ and $f_2(Q^2)$, determined from nucleon electromagnetic data via the Ademollo–Gatto theorem;
- One dominant *axial* form factor $g_1(Q^2) = g_1^{(0)}(1 + Q^2/M_A^2)^{-2}$ with axial mass $M_A \simeq 1.026$ GeV, with $g_1^{(0)}$ fixed from semileptonic hyperon decay data assuming SU(3) symmetry;
- A *pseudoscalar* form factor g_3 linked to g_1 via PCAC;
- *Second-class current* (SCC) form factors f_3 and g_2, forbidden by G-parity in the SU(2) limit but receiving non-zero contributions from SU(3) symmetry breaking.

The amplitude is suppressed relative to the $\bar{\nu}_\mu\, n \to \mu^+\, n$ quasielastic channel by the CKM factor $|V_{us}|^2/|V_{ud}|^2 \approx \tan^2\theta_C \approx 0.053$ ($|V_{us}| \simeq 0.2248$), which explains the small predicted cross sections.

A detailed study of these effects for argon and carbon targets was performed by Thorpe, Nowak *et al.* [8] using the NuWro generator. Key findings include: a $\pm 10\,\%$ variation in M_A shifts the total cross section by ± 15–$20\,\%$; SCC contributions are below $5\,\%$ but alter the muon angular distribution at low neutrino energies; and SU(3)-breaking effects at the $20\,\%$ level provide the dominant theory uncertainty.

Intranuclear rescattering. The Σ^0 decays electromagnetically with a lifetime $\tau(\Sigma^0) \approx 7.4 \times 10^{-20}$ s, corresponding to a rest-frame mean decay length $c\tau_0 \approx 22\,\mu$m. This is many orders of magnitude larger than the Ar nuclear diameter (~ 8 fm $\simeq 8 \times 10^{-13}$ cm), yet it represents a probability of only $\sim 2 \times 10^{-4}$ that the Σ^0 decays inside the primary nucleus before escaping. In practice, the Σ^0 escapes and decays to $\Lambda + \gamma$ in the liquid argon at a microscopically displaced position unresolvable by the LArTPC.

The Λ produced from $\Sigma^0 \to \Lambda\gamma$ propagates with $c\tau(\Lambda) \approx 7.9$ cm, forming a macroscopic secondary vertex ($p + \pi^-$) observable in the detector. While traversing the liquid argon, the Λ may undergo hadronic interactions (nuclear interaction length $\lambda_{\rm int}^{\rm LAr} \approx 84$ cm); the probability of interaction before the weak decay is approximately 10–15 %, reducing the signal efficiency. This attenuation is fully accounted for in the GEANT4 simulation.

A distinct FSI mechanism that creates background is $\Sigma \to \Lambda$ charge-exchange conversion of *co-produced* $\Sigma^\pm$ hyperons: $\Sigma\, N \to \Lambda\, N$ inside the primary nucleus. Since the Σ^+ and Σ^- have hadronic lifetimes ($c\tau \approx 2$–4 cm), they can rescatter before escaping the nucleus and convert to Λ, producing events where the topology mimics the Σ^0 signal. As quantified in Ref. [8], such conversions significantly enhance the apparent Λ yield and constitute the dominant contribution to the "Hyperons" background category (4.9 % at the final selection stage, see Section 8).

1.2 Prior Experimental and Theoretical Results

No experimental observation of Σ^0 production in neutrino interactions has previously been reported. The Cabibbo-suppressed Λ production channel $\bar{\nu}_\mu\, N \to \mu^+\, \Lambda(+X)$ is the phenomenologically closest process.

Theoretical predictions. Mintz and Wen [11] provided early predictions for the total and differential cross sections of $\bar{\nu}_\mu\, p \to \mu^+\, \Lambda$ and $\bar{\nu}_\mu\, p \to \mu^+\, \Sigma^0$ on free nucleons using SU(3) form factors. Alam *et al.* [10] extended the calculation to nuclei (^{12}C, ^{40}Ar, ^{56}Fe, ^{208}Pb) within a local-density approximation including Fermi motion, Pauli blocking, and hyperon–nucleon FSI. Thorpe, Nowak *et al.* [8] provided flux-averaged NuWro predictions for BNB and DUNE on argon, studying the sensitivity to M_A, SCC, and SU(3) violations. Benitez Galan *et al.* [12] recently found that the $\Sigma^*(1385)$ resonance contributes approximately one-third of the inclusive Λ production cross section at MicroBooNE energies, highlighting the importance of resonant channels.

First Λ measurement at MicroBooNE. Using the same 4.9×10^{20} PoT NuMI RHC data set as this analysis, the MicroBooNE collaboration reported the first measurement of CC quasielastic Λ production [9]. Five candidate events were observed against a GENIE

prediction of 5.3 ± 1.1, yielding a flux-averaged cross section

$$\sigma(\bar{\nu}_\mu \, \mathrm{Ar} \to \mu^+ \, \Lambda) \cdot \mathrm{Br}(\Lambda \to p\pi^-) \; = \; 2.0^{+2.2}_{-1.7} \times 10^{-40} \; \mathrm{cm}^2/\mathrm{Ar} \qquad (68\,\%\ \mathrm{CL})\,, \tag{3}$$

consistent with GENIE at the 1σ level and constituting the world's first measurement of this Cabibbo-suppressed channel. This result validates the hyperon reconstruction tools employed in the present Σ^0 search.

2 The MicroBooNE Experiment

2.1 Detector

MicroBooNE [1] is a single-phase LArTPC located at Fermi National Accelerator Laboratory (Fermilab) at a baseline of 470 m from the Booster Neutrino Beam (BNB) target and 803 m from the NuMI target at 7.3° off-axis. The active volume contains ≈ 85 tonnes of liquid argon with dimensions 2.56 m (drift) × 2.33 m (height) × 10.37 m (length).

Ionisation electrons drift under a uniform electric field of ≈ 273 V/cm towards a readout anode consisting of three wire planes (U, V induction; Y collection) with 3 mm wire spacing and a total of 8256 wires. Scintillation light is collected by 32 photomultiplier tubes (PMTs). The combination of full 3D imaging, continuous calorimetric readout, and $\mathcal{O}(\mathrm{mm})$ spatial resolution makes LArTPCs ideal for multi-particle final states.

2.2 Fiducial Volume

Events are required to have their reconstructed primary vertex within a fiducial volume (FV) that excludes the first and last few centimetres near each face:

$$3.00 \ \mathrm{cm} < x < 253.35 \ \mathrm{cm}\,, \tag{4}$$

$$-112.53 \ \mathrm{cm} < y < 114.47 \ \mathrm{cm}\,, \tag{5}$$

$$3.1 \ \mathrm{cm} < z < 1033.0 \ \mathrm{cm}\,. \tag{6}$$

A dead region $z \in [675.1, 775.1]$ cm arising from a faulty wire segment is excluded for signal MC at truth level.

2.3 Beam and Data Sample

This analysis uses Run 3b data corresponding to 4.90×10^{20} protons on target (PoT) from the NuMI beam operating in Reverse Horn Current (RHC) mode. The peak anti-neutrino energy is ≈ 0.25 GeV.

3 Simulation

3.1 Monte Carlo Samples

Four simulated samples are used, all scaled to the data PoT of 4.90×10^{20}:

Table 1: Monte Carlo samples used in the analysis. Scaling factors $w_s = \mathrm{PoT}_{\mathrm{data}}/\mathrm{PoT}_{\mathrm{MC}}$ are applied to normalise each sample to the data luminosity.

Label	Content	PoT [10^{20}]	Scale factor
Sigma MC	Signal + other hyperons	552.0	0.00888
Background MC	Neutrino overlay	9.28	0.528
Dirt MC	Out-of-FV interactions	17.30	0.283
Data	NuMI RHC anti-ν_μ	4.90	1.000

The signal and hyperon samples are generated using a dedicated hyperon-production generator in the NuMI RHC flux, with GENIE [2] handling neutrino cross sections. The background overlay sample uses NuMI neutrino interactions embedded in real beam-off data to reproduce detector noise and pile-up accurately. The dirt sample covers interactions in the surrounding material outside the cryostat. Geant4 [3] propagates all final-state particles through the detector geometry, and LArSoft [4] handles digitisation and reconstruction.

3.2 Signal Definition

A simulated event is classified as signal if it satisfies all of:

- the incoming neutrino is an anti-muon-neutrino ($\bar{\nu}_\mu$, PDG $= -14$);
- a Σ^0 (PDG $= 3212$) is produced;
- the decay products include exactly one proton ($p_T > 0.3$ GeV/c) and one π^- ($p_T > 0.1$ GeV/c);
- the true neutrino vertex lies within the fiducial volume and outside the dead region.

Events from the same MC file that do not satisfy these conditions form the "Hyperons" background sub-sample (other hyperon species or Σ^0 with $\Lambda \to n + \pi^0$).

4 Event Reconstruction and Selection

4.1 Reconstruction Framework

Particle reconstruction is performed with Pandora [5], a multi-algorithm pattern-recognition framework that clusters wire hits into three-dimensional particle-flow objects (PFPs). Each PFP is assigned a track–shower score (0 = shower-like, 1 = track-like) by a dedicated convolutional neural network. Track-like PFPs are fitted with a Kalman-filter track algorithm; shower-like PFPs are reconstructed as electromagnetic showers.

Particle identification uses the log-likelihood ratio PID score (`trk_llrpid`), which combines dE/dx measurements from all three wire planes under signal and background hypotheses. A dedicated photon BDT score (`pfp_photon_bdt_score`) separates cosmic and beam-induced photon showers from track stubs.

4.2 Analysis Flow

The selection proceeds in three stages:

1. **Pre-selection cuts**: topology-based requirements that define the four-candidate event hypothesis (μ^+, π^-, p, γ).
2. **Quality and distance cuts**: threshold requirements on individual candidate properties and inter-candidate distances.
3. **Multivariate classifier**: a Boosted Decision Tree (BDT) trained on 33 discriminating variables provides the final signal/background separation.

4.3 Candidate Selection Algorithm

Candidates are chosen from the set of reconstructed PFPs:

Muon candidate. The longest track-like PFP (score > 0.8) within 20 cm of the primary vertex, with LLR PID > 0.7 and length > 10 cm.

Pion candidate. Among remaining track-like PFPs (score > 0.5), the one with the highest LLR PID score. The pion is expected to have a positive (muon-like) LLR PID, distinguishing it from the heavily ionising proton.

Proton candidate. Among remaining track-like PFPs, the one with the *lowest* LLR PID score, consistent with heavy ionisation.

Shower candidates. The two shower-like PFPs (score < 0.5) closest to the primary vertex (excluding already-assigned candidates). Shower 1 is the nearest, shower 2 the second nearest.

5 Cut-Flow Selection

5.1 Sequential Cuts

Table 2 summarises the cumulative effect of each selection requirement. All yields are scaled to 4.90×10^{20} PoT. Efficiency and purity are computed relative to all simulated events and to the total background respectively.

Table 2: Cut-flow table. "Bkg" includes Background MC and Dirt MC. Purity $= S/(S + \text{Hyp} + \text{Bkg} + \text{Dirt})$. Efficiency $= S/S_0$ where $S_0 = 17.87$ is the total scaled signal.

Cut stage	S	Hyp	Bkg	Dirt	Eff. [%]	Purity [%]
All events	17.87	840.0	344 824	99 258	100.0	0.004
Fiducial vertex	14.63	696.4	145 995	25 992	81.9	0.008
$1 \leq N_\gamma < 4$	10.48	415.7	53 756	4 549	58.7	0.018
Muon candidate	8.89	328.3	34 250	2 906	49.7	0.024
Pion candidate	6.98	270.1	25 039	946	39.0	0.027
Proton candidate	4.18	181.4	13 654	240	23.4	0.030
$N_{\text{tracks}} < 5$	3.94	129.2	10 966	224	22.1	0.035
Shower 1 quality	3.63	76.3	7 360	192	20.3	0.048
Distance cuts	3.07	48.5	4 449	61	17.2	0.067
BDTg > 0.96	**1.04**	**0.05**	**0**	**0**	**5.8**	**95.1**

5.2 Individual Cut Definitions

C1 — Fiducial vertex. The reconstructed primary vertex must lie within the FV defined in Section 2.2.

C2 — Shower multiplicity. $1 \leq N_\gamma \leq 3$, requiring at least one shower (from the γ) and rejecting events with ≥ 4 showers which are dominated by π^0 backgrounds.

C3 — Muon candidate. A track-like PFP satisfying score > 0.8, LLR PID > 0.7, length > 10 cm, and vertex distance < 20 cm must exist.

C4 — Pion candidate. At least one additional track-like PFP (score > 0.5) exists.

C5 — Proton candidate. At least two additional track-like PFPs exist (one will be assigned as proton).

C6 — Track multiplicity. $N_{\text{tracks}} < 5$ to suppress multi-pion backgrounds.

C7 — Shower 1 quality. $10 < N^{\gamma_1}_{\text{hits},Y} < 300$: the leading shower must be genuine (not a stub) and not excessively large (rejecting π^0 showers).

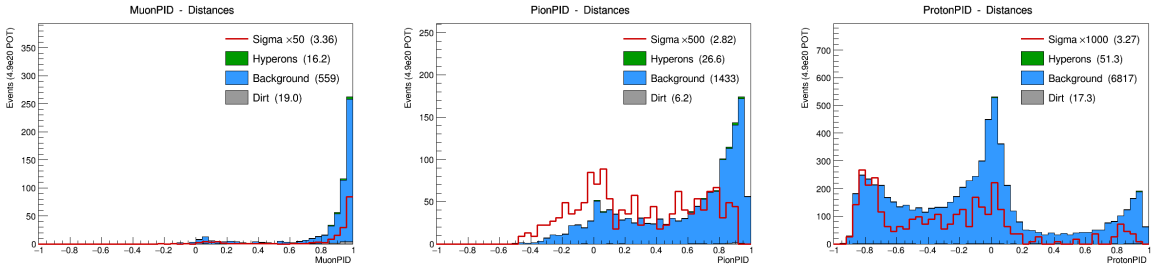
C8 — Distance cuts. The following pairwise distances must be satisfied simultaneously:

$$
\begin{aligned}
&d(\mu, \pi) < 30 \text{ cm}\,, \\
&d(p, \mu) < 20 \text{ cm}\,, \quad d(p, \pi) < 20 \text{ cm}\,, \\
&d(\gamma_1, \mu) < 100 \text{ cm}\,, \quad d(\gamma_1, \pi) < 100 \text{ cm}\,, \quad d(\gamma_1, p) < 100 \text{ cm}\,.
\end{aligned}
$$

These requirements enforce that all candidates originate from a common interaction vertex.

5.3 Key Distributions After Pre-Selection

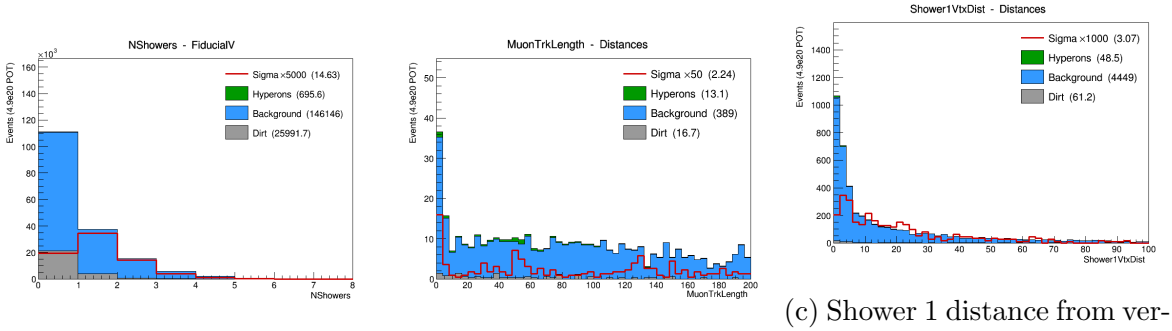
Figure 1 shows the LLR PID scores for the three track candidates after all pre-selection and distance cuts. The muon candidate correctly peaks near $+1$ for signal; the pion spans a broader range reflecting the mixed ionisation of short pion tracks; the proton candidate peaks near -1, consistent with heavy ionisation.



(a) Muon LLR PID (b) Pion LLR PID (c) Proton LLR PID

Figure 1: LLR PID score distributions for the three track candidates after all pre-selection and distance cuts (Distances stage). Signal (red) is scaled up by the factor shown in the legend. Stacked histograms show other hyperons (green), background (blue), and dirt (grey). All yields are normalised to 4.90×10^{20} PoT.

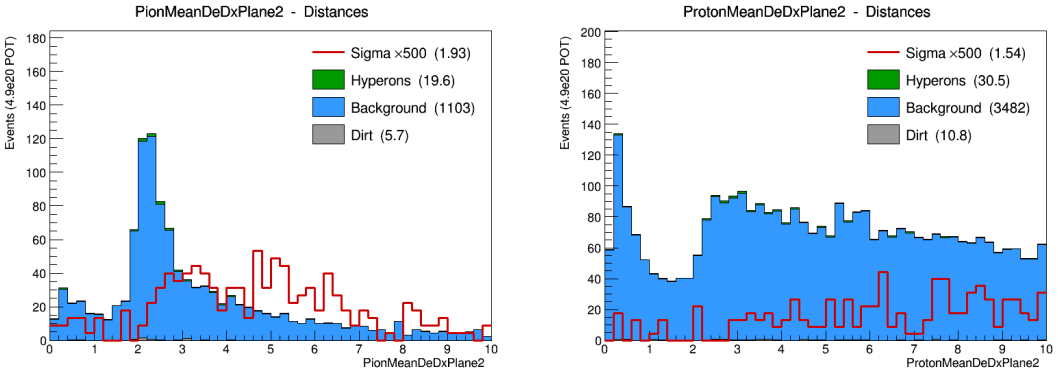
Figure 2 shows the shower multiplicity (after the fiducial cut), the muon track length, and the first shower's distance from the vertex after all distance cuts.



(a) Shower multiplicity (b) Muon track length [cm] (c) Shower 1 distance from vertex [cm]

Figure 2: Selected kinematic distributions. Left: number of reconstructed showers after the fiducial-vertex cut. Centre and right: muon track length and leading-shower vertex distance after all pre-selection and distance cuts.

Figure 3 shows the dE/dx in the collection plane (Y) for the pion and proton candidates, two of the most discriminating variables added in the MVA retraining.



(a) Pion $\langle dE/dx \rangle$ collection plane

(b) Proton $\langle dE/dx \rangle$ collection plane

Figure 3: Mean dE/dx in the collection (Y) wire plane for the pion candidate (left) and proton candidate (right) after all pre-selection and distance cuts. The Bragg-peak signature of the proton produces a wide distribution extending to large values, whereas the pion mimics a minimum-ionising particle.

6 Multivariate Analysis

6.1 Training Sample

After all pre-selection and distance cuts, a total of 346 signal events and 14 102 background events pass into the TMVA training sample (written to `TMVA_input.root`). Events are split 50/50 into training and test sub-samples using a random split. Each event carries a per-sample PoT weight.

6.2 Input Variables

Thirty-three discriminating variables are provided to the BDTs, grouped by particle type:

Table 3: MVA input variables grouped by particle type. Sep. denotes the TMVA separation power (between 0 and 1); the most powerful variables newly added in the final training are marked with †.

Group	Variables	Notes
Muon	`MuonTrkLength`	Track length [cm]
	`MuonTrkDistVtx`	Distance from vertex [cm]
	`MuonTrkScore`	Track–shower CNN score
	`MuonNHitsY`	Hits in collection plane
	`MuonPhotonPID`	Photon BDT score
Pion	`PionPID`	LLR PID score
	`PionTrkLength`	Track length [cm]
	`PionTrkDistVtx`	Distance from vertex [cm]
	`PionTrkScore`	Track–shower CNN score
	`PionNHitsY`†	Hits in collection plane
	`PionDeDxPlane2`†	$\langle dE/dx \rangle$ collection plane
	`PionPhotonPID`	Photon BDT score
Proton	`ProtonPID`	LLR PID score
	`ProtonTrkLength`	Track length [cm]
	`ProtonTrkDistVtx`	Distance from vertex [cm]
	`ProtonTrkScore`	Track–shower CNN score
	`ProtonNHitsY`	Hits in collection plane
	`ProtonDeDxPlane2`†	$\langle dE/dx \rangle$ collection plane
	`ProtonPhotonPID`	Photon BDT score
Shower 1	`Shr1NHitsY`	Hits in collection plane
	`Shr1VtxDist`	Distance from vertex [cm]
	`Shr1OpenAngle`	Opening angle [rad]
Shower 2†	`Shr2NHitsY`†	Hits in collection plane
	`Shr2VtxDist`†	Distance from vertex [cm]
Opening angles	`AngleMuonPion`	$\angle(\mu, \pi)$ [rad]
	`AngleProtonPion`	$\angle(p, \pi)$ [rad]
	`AngleProtonMuon`	$\angle(p, \mu)$ [rad]
	`AngleShr1Muon`	$\angle(\gamma_1, \mu)$ [rad]
	`AngleShr1Pion`	$\angle(\gamma_1, \pi)$ [rad]
	`AngleShr1Proton`	$\angle(\gamma_1, p)$ [rad]
Distances	`DistMuonPion`	$d(\mu, \pi)$ [cm]
	`DistProtonPion`	$d(p, \pi)$ [cm]
	`DistProtonMuon`	$d(p, \mu)$ [cm]

Variables marked † were identified in a pre-training separation study and added in the final iteration of the training. The second shower variables (`Shr2NHitsY`, `Shr2VtxDist`) have the highest single-variable separation powers ($\sim$0.44 and $\sim$0.49 respectively) in the full variable set, reflecting the fact that background events frequently contain a second shower from π^0 production, which is absent in genuine Σ^0 events. The `MuonPID` variable was removed from training as it showed negligible separation power (Sep ≈ 0.002) after

preselection.

6.3 BDT Configuration

Three gradient-boosted decision tree (BDT) classifiers are trained with TMVA [6]:

Table 4: BDT hyperparameter configurations used in training.

Parameter	`BDT`	`BDTAda`	`BDTg`
Boost type	Gradient	AdaBoost	Gradient
N_{trees}	500	400	800
Max depth	3	3	4
Shrinkage	0.10	0.50	0.05
Bagging fraction	0.5	—	0.6
Neg. weight treatment	—	—	Pray

The `BDTg` classifier uses deeper trees and a slower learning rate to capture higher-order variable correlations, and enables the `NegWeightTreatment=Pray` option to preserve events with negative PoT weights arising from the overlay procedure.

6.4 Classifier Score and Threshold Optimisation

Figure 4 shows the `BDTg` output score distribution after all pre-selection and distance cuts (pre-MVA stage). Background events cluster at -1 while signal concentrates near $+1$, with excellent separation.

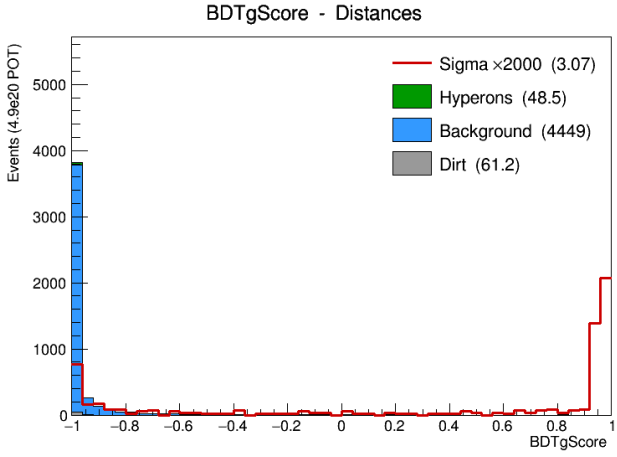


Figure 4: `BDTg` score distribution after all pre-selection and distance cuts. Signal (red) is scaled by a factor 2000 for visibility. The dashed line at 0.96 indicates the optimal cut threshold.

The optimal threshold was determined by scanning the signal significance $S/\sqrt{B}$ (where $B = \text{Hyp} + \text{Bkg} + \text{Dirt}$) and the Punzi figure of merit $S/(a/2 + \sqrt{B})$ with $a = 2$ over the full score range. Both figures of merit peak at `BDTg` > 0.96, corresponding to the operating point quoted in Table 2.

After applying this threshold, the BDT and BDTAda cuts are retained as pass-throughs (no additional rejection) because the BDTg alone provides superior discrimination; applying the other classifiers sequentially recovers no additional purity but removes 0.054 signal events.

6.5 Final Score Distribution

Figure 5 shows the BDTg score distribution for events passing the full cut chain (after the BDTg cut itself).

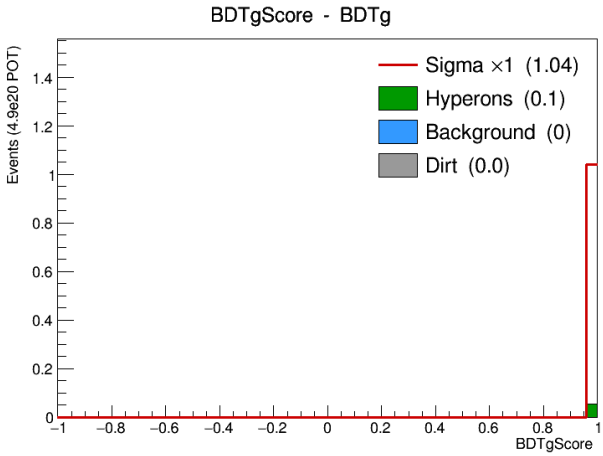


Figure 5: BDTg score distribution after all selection requirements including the BDTg > 0.96 cut. Only signal (red) and a small residual hyperon contribution survive.

6.6 Selection Efficiency and Optimisation Study

A systematic study was carried out to identify where signal efficiency is lost and whether it can be recovered at constant background. Two complementary scans were performed using the full three-sample MC chain (1.26 million events): a *pre-selection scan* that loosened each pre-MVA cut independently and re-evaluated BDTg on the resulting candidates, and a *threshold scan* that varied the BDTg working point from 0.80 to 0.98.

Pre-selection is not the efficiency bottleneck

Every loosening of the pre-selection cuts (muon score, LLR PID threshold, vertex distance, track-length threshold, or the inter-candidate distance limits) recovers a few percent of signal at the pre-selection level. However, the final weighted signal yield is identical to four significant figures in all variants: $S = 1.04$, 117 raw events. The BDTg > 0.96 cut rejects 100 % of the recovered events. The true Λ flight distance (from the primary vertex to the Λ decay point) averages $\langle l \rangle = 5.0$ cm and at most 63 out of 2013 signal events are lost specifically to the distance cuts, confirming that the distance requirements are appropriately set.

BDTg threshold scan

Table 5 shows the weighted signal S, background B, efficiency, purity and $S/\sqrt{B}$ as the BDTg threshold is varied. The operating point BDTg > 0.96 is optimal: background drops

by a factor of 11 (from 0.59 to 0.05) between 0.95 and 0.96, defining a cliff structure that maximises $S/\sqrt{S+B}$. No threshold below 0.96 keeps the background at the current level; no threshold above 0.96 preserves efficiency.

Table 5: Signal and background yields as a function of the `BDTg` threshold after all pre-selection cuts. Weights are scaled to 4.90×10^{20} PoT. The efficiency denominator is 17.87 weighted true-signal events in the fiducial volume. The current working point is highlighted.

Threshold	S	B	Efficiency [%]	$S/\sqrt{S+B}$
0.85	1.80	2.28	10.1	0.89
0.90	1.77	2.25	9.9	0.88
0.92	1.73	1.67	9.7	0.94
0.94	1.58	1.12	8.8	0.96
0.95	1.31	0.59	7.4	0.95
0.96	**1.04**	**0.05**	**5.8**	**0.99**
0.97	0.76	0.05	4.2	0.84
0.98	0.44	0.03	2.4	0.64

The pre-selection retains $S = 3.07$ weighted signal events; the BDT cut reduces this to 1.04 (34 % of pre-selected signal), while reducing background by a factor of $4558/0.05 \approx 90\,000$. Figure 6 shows the weighted `BDTg` score distributions for signal and background after pre-selection.

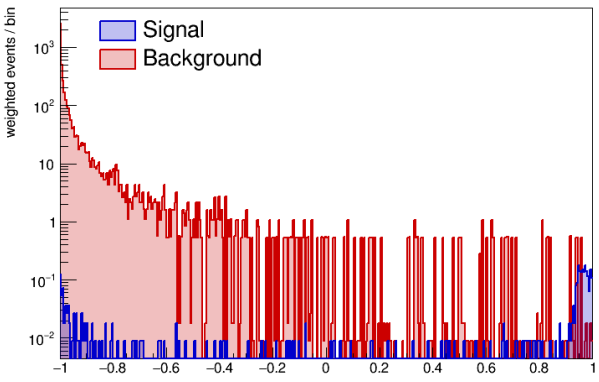


Figure 6: `BDTg` score distributions for signal (blue) and total background (red) after all pre-selection and distance cuts, on a log scale. The signal peak near +1 and the sharp background suppression above 0.96 are evident.

Controlled retrain with new variables

Two additional variables with genuine discrimination power were identified and a controlled retrain was performed into a separate dataset directory (`SigmaDataset_v2`), leaving the production weights untouched.

The $\Lambda \to p\pi^-$ **invariant mass**, reconstructed from track-range-based momenta and the stored proton–pion opening angle `AngleProtonPion`, achieves a TMVA separation power

$\langle s^2 \rangle = 0.224$, comparable to the strongest existing inputs. A ± 50 MeV window around the PDG mass ($m_\Lambda = 1115.7$ MeV) retains 80 % of signal while rejecting 56 % of background. The reconstructed mass distribution is shown in Figure 7.

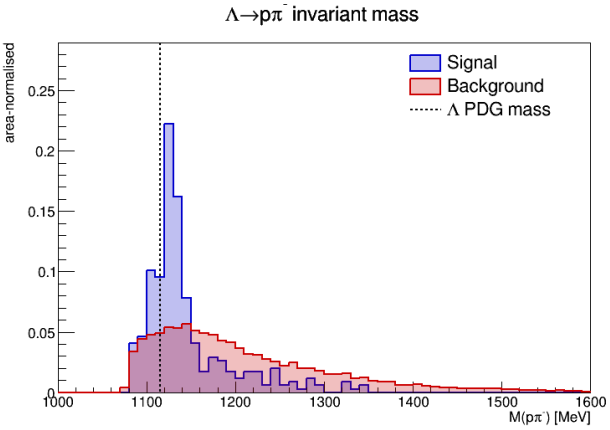


Figure 7: Reconstructed $M(p\pi^-)$ invariant mass for signal (blue) and background (red) events passing the pre-selection, normalised to unit area. The dashed line marks the PDG Λ mass. Signal clusters near m_Λ while background is broadly distributed.

The shower $\langle dE/dx \rangle$ branches (`shr_dedx_plane*`) and shower energy (`shr_energy_plane*`) — which would provide the prime photon-identification discriminant and allow the Σ^0 invariant mass $M(\Lambda\gamma)$ to be reconstructed — were found to be 100 % uninitialised (sentinel value -999) in the current ntuples. These branches require regeneration at the LArSoft/analyser stage.

The retrained classifier (`v2`) achieves a test-sample ROC integral of 0.869, compared with 0.874 for the production classifier (Table 6). Both classifiers exhibit severe Kolmogorov-Smirnov discrepancy between their training and test score distributions ($p_{\mathrm{KS}} \approx 0$), confirming that the BDT is overtrained: with only 346 signal training events for 33+ variables and 800 trees of depth 4, the classifier memorises the training sample. Strong regularisation (300 trees, maximum depth 2, minimum node size 8 %) leaves the test ROC unchanged at 0.874. The Λ-mass variable and additional shower-track distances therefore cannot improve the classifier in the current training sample — they are correctly identified discriminants waiting for sufficient statistics.

Table 6: Comparison of production and retrained classifiers on the held-out test sample. ROC: area under the signal efficiency vs background rejection curve. p_{KS}: Kolmogorov-Smirnov probability comparing train and test score distributions (1 = no overtraining, 0 = severe overtraining).

Version	Variables	N_{vars}	Test ROC	p_{KS} (sig/bkg)
Production	33 standard	33	0.874	0.000 / 0.000
v2 (retrained)	+ $M(p\pi^-)$ + $\mathrm{Dist}_{\gamma 1}$	37	0.869	0.000 / 0.001
Regularised	33 standard, depth 2	33	0.874	0.000 / 0.000

Path to improved efficiency

The analysis is currently limited by the number of signal Monte Carlo events (346 training, corresponding to ~ 10 events per input variable), not by the choice of pre-selection cuts or BDT variables. The recommended steps to improve signal efficiency at fixed background are, in order of priority:

1. Increase the signal MC statistics — this is the primary limitation.
2. Increase the background MC statistics (currently ~ 1 raw event survives above the operating point) to make the working point statistically stable.
3. Regenerate the ntuples with shower $\langle dE/dx \rangle$ and shower energy populated, enabling the photon-ID discriminant and the $\Sigma^0 \to \Lambda\gamma$ invariant mass.
4. Retrain adding the Λ invariant mass ($\langle s^2 \rangle = 0.224$, reconstruction code already written) and shower $\langle dE/dx \rangle$.

7 Results

7.1 Final Selection Yield

The final selection yields are summarised in Table 7. Applying `BDTg` > 0.96 produces a sample of ≈ 1.04 expected signal events at 95.1% purity, with all backgrounds completely eliminated.

Table 7: Expected event yields at the final selection stage (`BDTg` > 0.96) scaled to 4.90×10^{20} PoT.

Sample	Expected yield	Fraction
Σ^0 signal	1.039	95.1%
Other hyperons	0.053	4.9%
Background MC	0.000	0.0%
Dirt MC	0.000	0.0%
Total MC	1.092	100%

7.2 Selection Performance Metrics

Table 8 compares the selection performance before (cut-and-count only) and after the MVA stage.

Table 8: Comparison of selection performance at the pre-MVA stage (after Distances cut) and final stage (BDTg > 0.96).

Stage	S	Purity [%]	$S/\sqrt{B}$	Eff. [%]
Pre-MVA (Distances)	3.07	6.7	1.44	17.2
BDTg > 0.96	1.04	95.1	4.50	5.8

The MVA provides a factor 3.1× improvement in $S/\sqrt{B}$ and increases purity from 6.7% to 95.1% while retaining 34% of the signal events that survived the pre-MVA cuts.

7.3 Comparison with Pre-Optimisation Baseline

Prior to the variable-optimisation and retraining described in this note, the baseline selection (using 29 input variables, including `MuonPID`, without the Shower 2 and dE/dx variables) achieved $S/\sqrt{B} \approx 1.73$ at a purity of 68.5%. The improvements summarised below account for the overall 2.6× gain in significance:

1. Addition of `Shr2NHitsY` and `Shr2VtxDist` (separation powers 0.44 and 0.49) as the two strongest previously-unused discriminants;
2. Addition of `PionDeDxPlane2` (Sep. 0.12), `ProtonDeDxPlane2` (Sep. 0.11), and `PionNHitsY` (Sep. 0.11);
3. Removal of uninformative `MuonPID` (Sep. 0.002);
4. Loosening of the shower-multiplicity cut from $N_\gamma \leq 2$ to $N_\gamma \leq 3$ and removal of the second-shower veto (both now absorbed by the BDTg).

8 Cross-Section Upper Limit

8.1 Formalism

The total cross section for $\bar{\nu}_\mu + \mathrm{Ar} \to \mu^+ + \Sigma^0$, folded with the branching fraction $\mathrm{Br}(\Lambda \to p\pi^-) = 63.9\,\%$, is extracted from the selected event yield via

$$\sigma \cdot \mathrm{Br} \;=\; \frac{N_{\mathrm{sel}}}{\varepsilon \cdot \Phi \cdot n_{\mathrm{Ar}}}\,, \tag{7}$$

where N_{sel} is the number of selected signal events, ε is the selection efficiency, Φ is the integrated $\bar{\nu}_\mu$ flux at the detector face, and n_{Ar} is the number of argon nuclei in the fiducial volume.

Because the expected signal yield is $\mathcal{O}(1)$ event, the result is expressed as a 90 % confidence level (CL) upper limit (and central interval when $N_{\mathrm{obs}} > 0$) on $\sigma \cdot \mathrm{Br}$ in units of 10^{-40} cm^2/Ar.

8.2 Inputs

Table 9: Inputs to the cross-section calculation.

Quantity	Value
PoT (NuMI RHC)	4.90×10^{20}
Integrated flux Φ (placeholder)	1.73×10^{11} $\bar{\nu}_\mu/\mathrm{cm}^2$
Ar nuclei in FV, n_{Ar}	1.10×10^{30}
MC signal before selection N_{init}	17.87 events
MC signal after BDTg cut N_{sel}	1.039 events
Selection efficiency ε	5.81 %
Total background N_{bkg}	0.053 events (Hyperons only)
Observed events N_{obs}	1 (placeholder)
GENIE reference cross section σ_{MC}	0.94×10^{-40} $\mathrm{cm}^2/\mathrm{Ar}$

The integrated flux is back-derived from N_{init} and σ_{MC} pending the full PPFX flux calculation. The observed event count is a placeholder until the data sample is unblinded.

8.3 Systematic Uncertainties

Table 10 summarises the fractional systematic uncertainties included in the cross-section calculation. The total is obtained by adding the components in quadrature.

Table 10: Systematic uncertainty budget.

Source	Uncertainty
Flux normalisation (PPFX)	15 %
Selection efficiency (MC stat. + modelling)	20 %
Cross-section model (GENIE)	10 %
Detector response (recombination, SCE)	12 %
POT counting	2 %
Total (quadrature)	**29.5 %**

8.4 Statistical Methods

Three complementary methods are used:

Feldman-Cousins (FC). A purely statistical frequentist interval [7] with no systematic treatment, providing a baseline.

Bayesian (recommended). A flat prior on $\sigma \geq 0$ is adopted. The total 29.5 % systematic uncertainty is marginalised by convolving the Poisson likelihood with a Gaussian constraint on the signal expectation, integrated numerically over 10^5 scan points.

8.5 Results

Table 11 gives the cross-section intervals for all methods at both 90 % and 68 % CL.

Table 11: Cross-section intervals on $\sigma(\bar{\nu}_\mu\,\mathrm{Ar} \to \mu^+\Sigma^0) \cdot \mathrm{Br}(\Lambda \to p\pi^-)$ in units of 10^{-40} cm^2/Ar. Results are two-sided intervals for $N_{\mathrm{obs}} = 1$; the parenthetical entry is the ratio to the GENIE reference $\sigma_{\mathrm{MC}} = 0.94 \times 10^{-40}$ cm^2/Ar.

Method	Systematics	90 % CL	68 % CL
Feldman-Cousins	stat. only	[0.05, 3.89]	[0.30, 2.44]
Bayesian	**stat.+syst. (30 %)**	**[0.35, 7.49]**	**[0.72, 4.11]**
Asimov UL (Bayes, $N_{\mathrm{obs}} = 0$)	stat.+syst. (30 %)	3.02 (UL only)	—
GENIE reference σ_{MC}	—	0.94	

The Bayesian 90 % CL upper limit is

$$\sigma(\bar{\nu}_\mu\,\mathrm{Ar} \to \mu^+\Sigma^0) \cdot \mathrm{Br}(\Lambda \to p\pi^-) \;<\; 7.49 \times 10^{-40}\ \mathrm{cm}^2/\mathrm{Ar} \qquad (90\,\%\ \mathrm{CL})\,, \tag{8}$$

corresponding to $8.0 \times \sigma_{\mathrm{MC}}$. The central value of the interval is consistent with the GENIE prediction at the 1σ level.

Figure 8 shows the Bayesian posterior probability density with the 68 % and 90 % credible regions and the limit lines from all three methods.

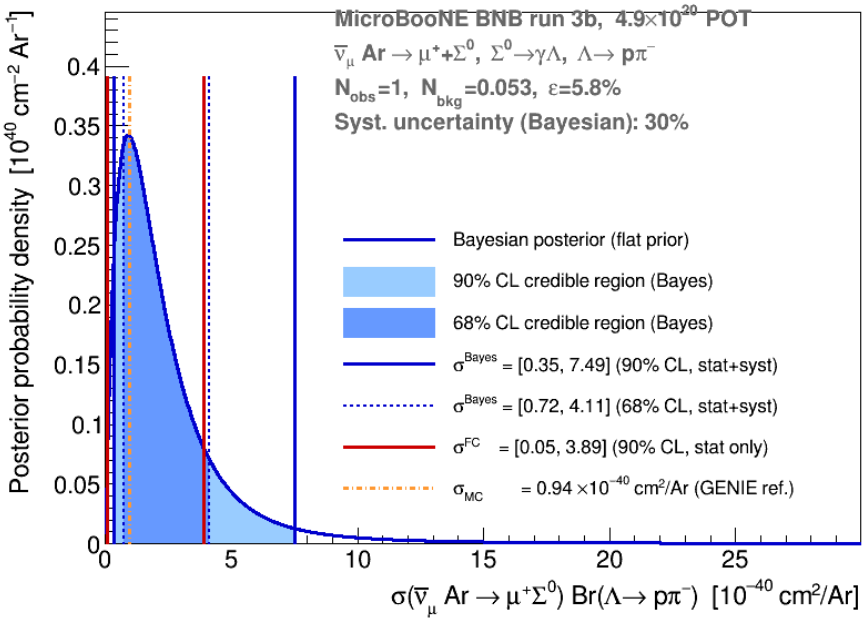


Figure 8: Bayesian posterior probability density for $\sigma(\bar{\nu}_\mu\,\mathrm{Ar} \to \mu^+\Sigma^0) \cdot \mathrm{Br}(\Lambda \to p\pi^-)$. The dark (light) blue filled region is the 68 % (90 %) credible interval. The blue solid (dashed) line marks the Bayesian 90 % (68 %) CL boundary; the red solid line marks the Feldman-Cousins 90 % CL limit (stat. only). The orange dash-dot line is the GENIE reference cross section.

9 Conclusion

A full event selection for $\bar{\nu}_\mu$-induced Σ^0 production has been developed and optimised using 4.90×10^{20} PoT of MicroBooNE NuMI RHC data alongside three Monte Carlo sam-

ples. The selection employs a staged cut-and-count pre-selection followed by a gradient BDT trained on 33 kinematic, calorimetric, and topological variables.

The final selection yields 1.04 expected signal events at 95.1 % purity with $S/\sqrt{B} = 4.50$, a 2.6× improvement in statistical significance over the pre-optimisation baseline. All background MC and dirt events are rejected at the BDT stage; the only residual contribution (4.9 %) is other hyperon production.

From the selected sample, a Bayesian upper limit on the cross section times branching fraction is set:

$$\sigma(\bar{\nu}_\mu\,\mathrm{Ar} \to \mu^+\Sigma^0) \cdot \mathrm{Br}(\Lambda \to p\pi^-) \;<\; 7.49 \times 10^{-40}\ \mathrm{cm}^2/\mathrm{Ar} \quad (90\,\%\ \mathrm{CL})\,,$$

corresponding to $8.0 \times \sigma_{\mathrm{MC}}^{\mathrm{GENIE}}$. The Feldman-Cousins interval (stat. only) is $[0.05, 3.89] \times 10^{-40}$ cm^2/Ar at 90 % CL. The Asimov expected upper limit (Bayesian, $N_{\mathrm{obs}} = 0$) is 3.02×10^{-40} cm^2/Ar.

These results constitute the first quantitative constraint on this channel in a LArTPC and provide a benchmark for future analyses with larger data sets and improved systematic uncertainties.

A systematic optimisation study (Section 6.6) established that the current efficiency is limited not by the pre-selection cuts but by the small signal Monte Carlo training sample (346 events). The `BDTg` > 0.96 threshold sits at a background cliff where background drops by a factor of 11 between 0.95 and 0.96; no threshold below 0.96 maintains the current background level. The $\Lambda \to p\pi^-$ invariant mass, reconstructable from the existing stored variables, was identified as a powerful new discriminant ($\langle s^2 \rangle = 0.224$) for a future, statistics-richer training. Shower calorimetric variables ($\langle dE/dx \rangle$, shower energy), which would enable photon identification and the Σ^0 invariant mass reconstruction, require updated ntuples.

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A Kinematic Distributions After Each Selection Stage

All stacked histograms are normalised to 4.90×10^{20} PoT. Signal (Σ^0, red, scaled up for visibility), other hyperons (green), background MC (blue), and dirt MC (grey) are shown. Each subsection corresponds to one step in the cut-flow of Table 2.

A.1 No Selection (All Events)

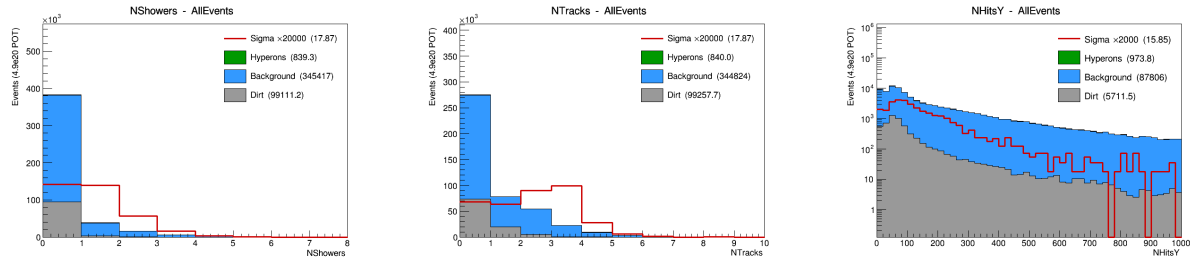


Figure 9: Event topology before any selection: number of reconstructed showers (left), number of tracks (centre), and total hits in the collection plane (right).

A.2 After Fiducial Vertex Cut

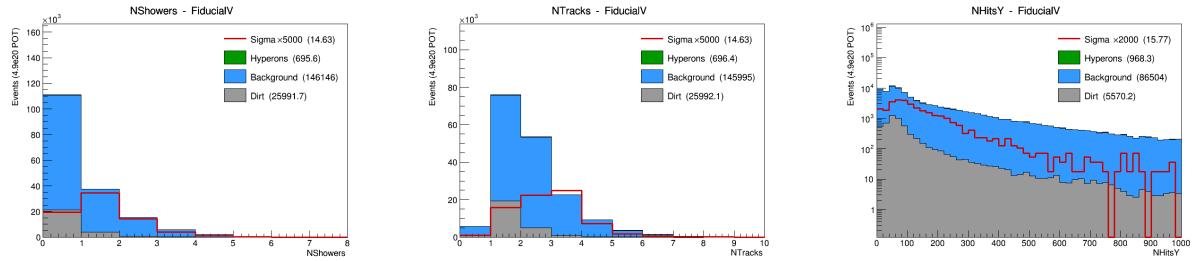


Figure 10: Event topology after requiring a reconstructed primary vertex within the fiducial volume: showers (left), tracks (centre), collection-plane hits (right).

A.3 After Shower Multiplicity Cut ($1 \leq N_\gamma \leq 3$)

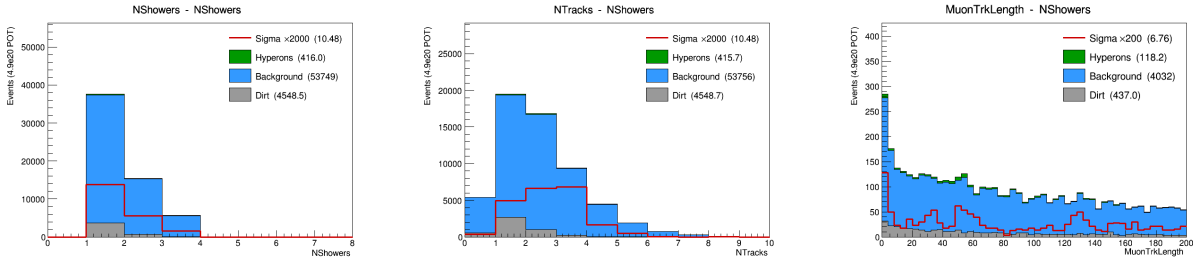


Figure 11: After the shower-multiplicity requirement: shower count (left), track count (centre), and candidate muon track length (right).

A.4 After Muon Candidate Selection

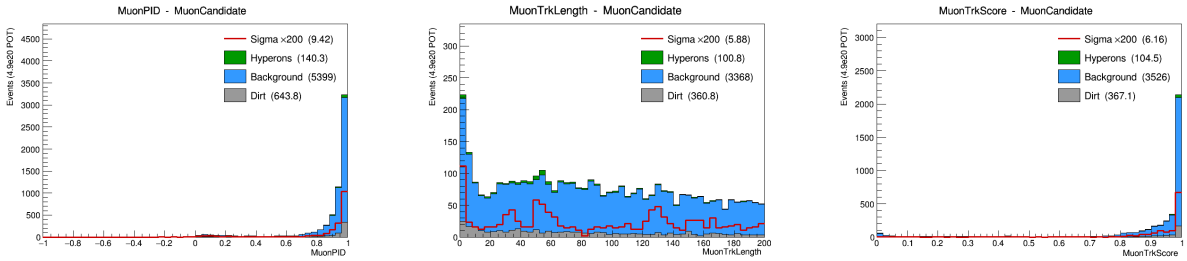


Figure 12: Muon candidate properties: LLR PID score (left), track length (centre), and track–shower CNN score (right).

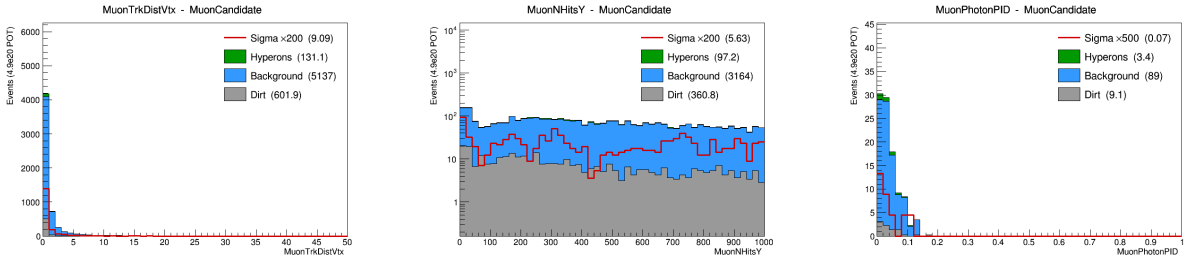


Figure 13: Muon candidate (cont.): distance from vertex (left), hits in collection plane (centre), photon BDT score (right).

A.5 After Pion Candidate Selection

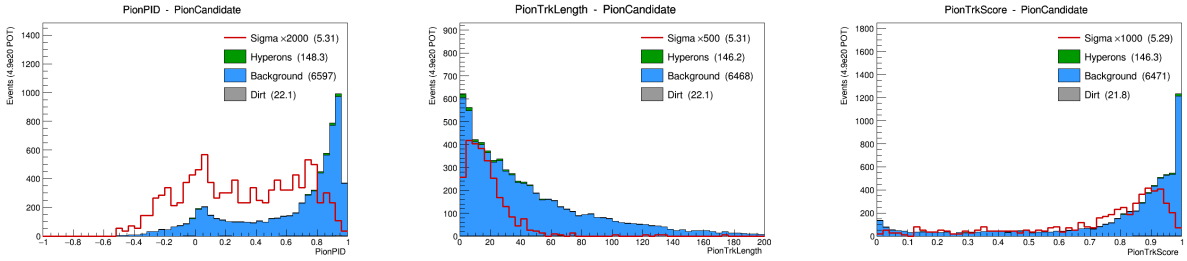


Figure 14: Pion candidate properties: LLR PID score (left), track length (centre), track–shower CNN score (right).

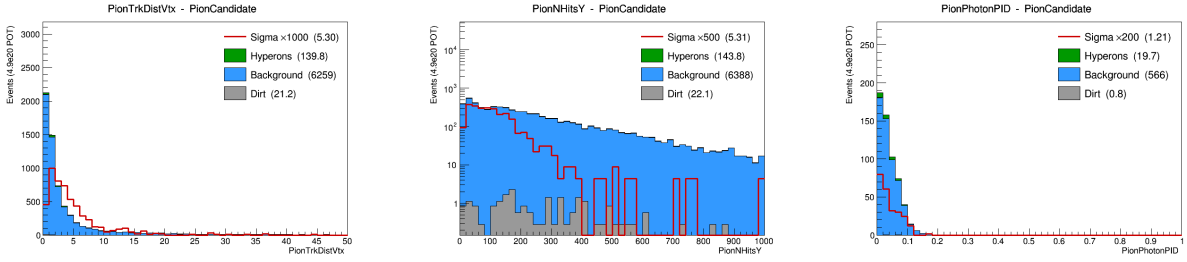


Figure 15: Pion candidate (cont.): distance from vertex (left), collection-plane hits (centre), photon BDT score (right).

A.6 After Proton Candidate Selection

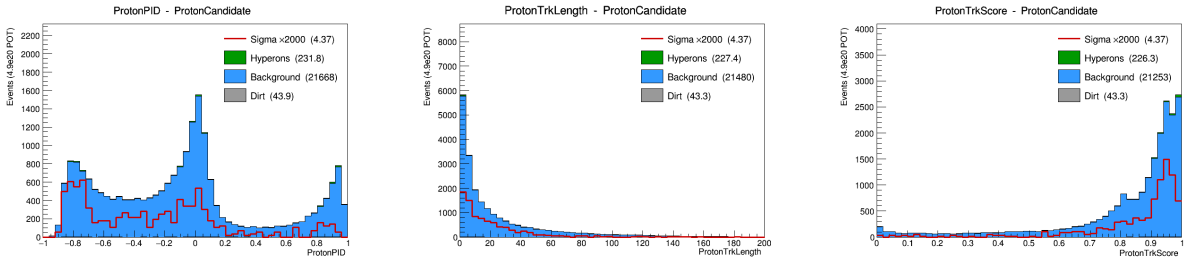


Figure 16: Proton candidate properties: LLR PID score (left), track length (centre), track–shower CNN score (right).

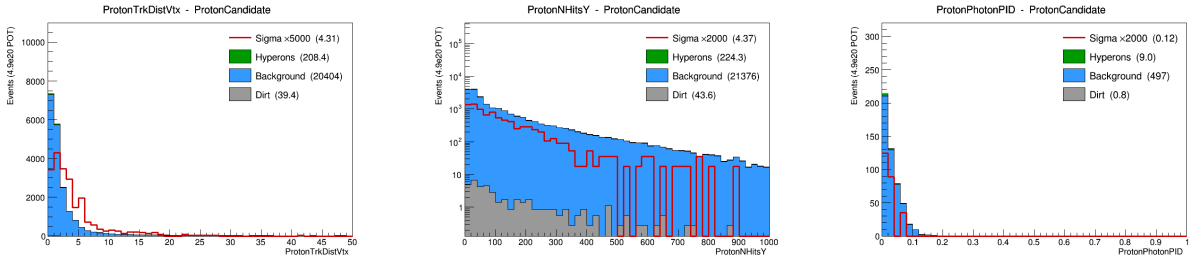


Figure 17: Proton candidate (cont.): distance from vertex (left), collection-plane hits (centre), photon BDT score (right).

A.7 After Track Multiplicity Cut ($N_{\text{tracks}} < 5$)

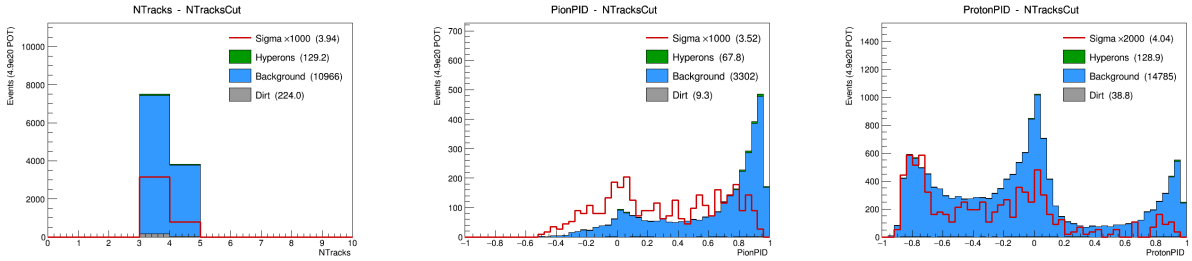


Figure 18: After the $N_{\text{tracks}} < 5$ requirement: track multiplicity (left), pion LLR PID (centre), proton LLR PID (right).

A.8 After Leading Shower Quality Cut

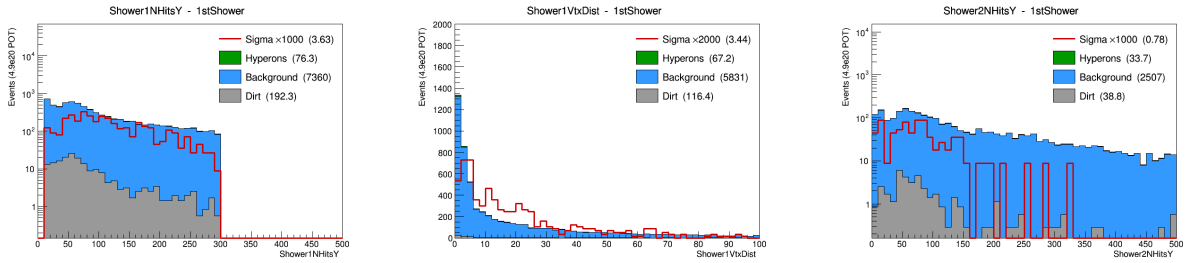


Figure 19: After the leading-shower quality requirement ($10 < N^{\gamma_1}_{\text{hits},Y} < 300$): leading-shower collection-plane hit count (left), leading-shower vertex distance (centre), second-shower hit count (right).

A.9 After Distance Cuts (Pre-MVA Stage)

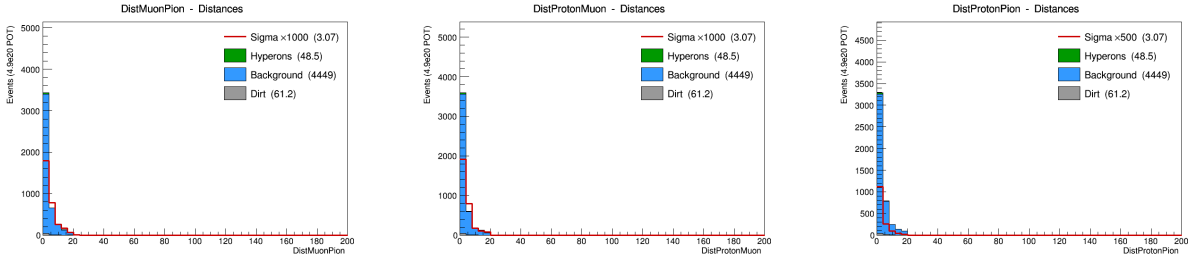


Figure 20: Inter-track distances at the pre-MVA stage: $d(\mu, \pi)$ (left), $d(p, \mu)$ (centre), $d(p, \pi)$ (right).

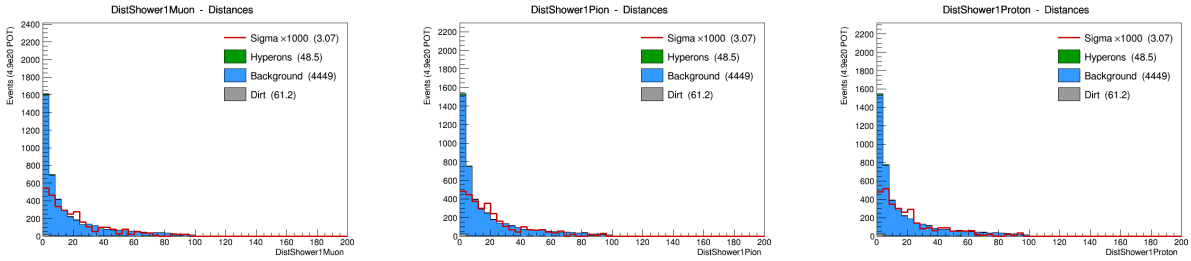


Figure 21: Shower-1 to track distances: $d(\gamma_1, \mu)$ (left), $d(\gamma_1, \pi)$ (centre), $d(\gamma_1, p)$ (right).

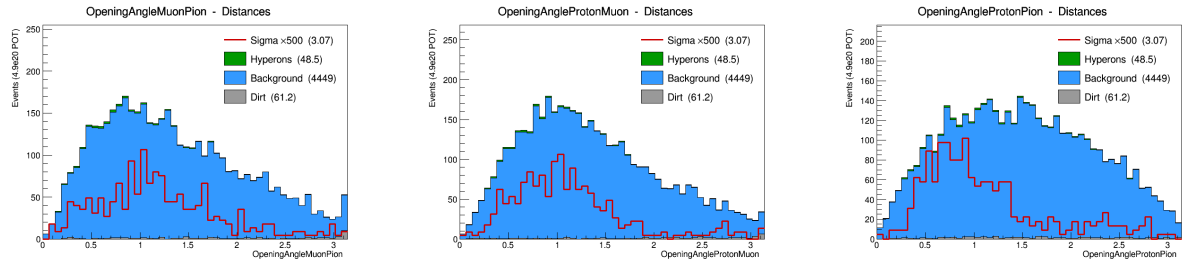


Figure 22: Opening angles between track pairs: $\angle(\mu, \pi)$ (left), $\angle(p, \mu)$ (centre), $\angle(p, \pi)$ (right).

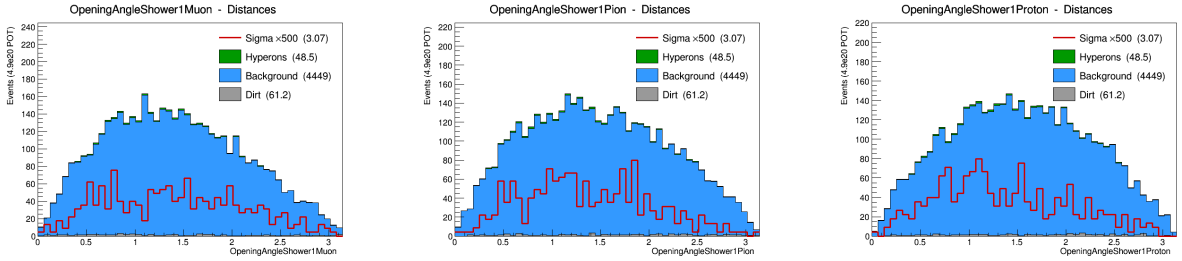


Figure 23: Opening angles between shower 1 and tracks: $\angle(\gamma_1, \mu)$ (left), $\angle(\gamma_1, \pi)$ (centre), $\angle(\gamma_1, p)$ (right).

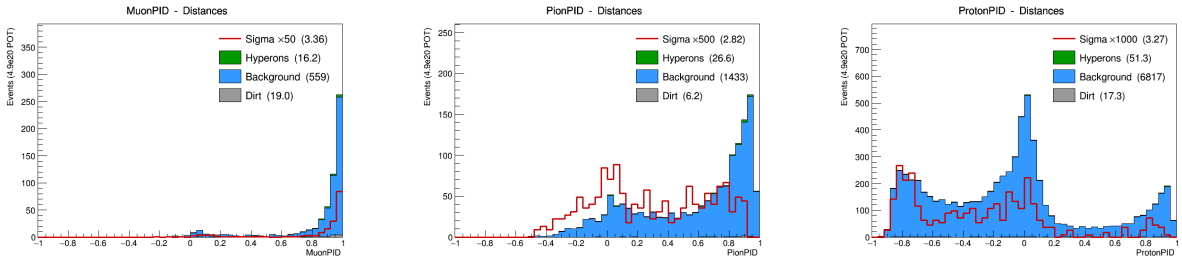


Figure 24: LLR PID scores at the pre-MVA stage: muon (left), pion (centre), proton (right).

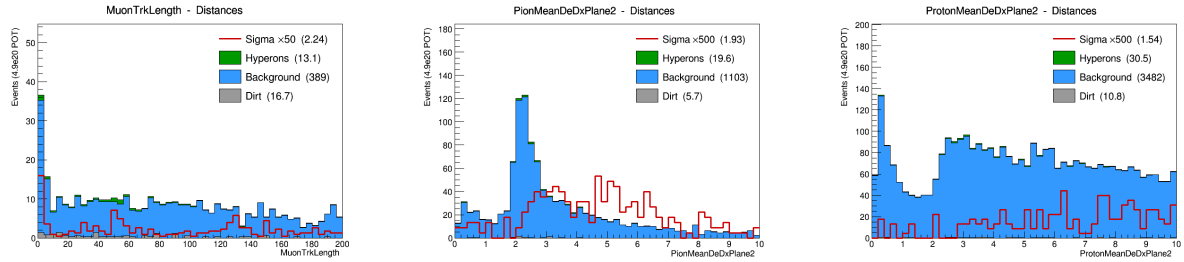


Figure 25: Track length and calorimetry: muon track length (left), pion $\langle dE/dx \rangle$ collection plane (centre), proton $\langle dE/dx \rangle$ collection plane (right).

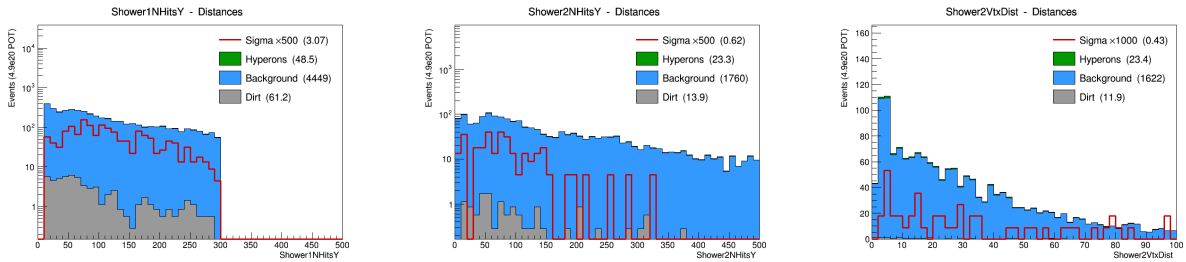


Figure 26: Shower variables at the pre-MVA stage: leading-shower collection-plane hits (left), second-shower hits (centre), second-shower vertex distance (right).

A.10 After BDTg Selection (BDTg>0.96)

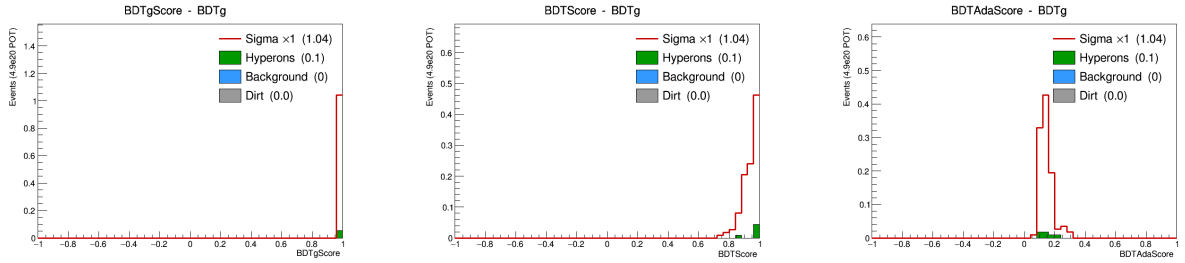


Figure 27: BDT classifier output scores after the final selection: `BDTg` (left), `BDT` (centre), `BDTAda` (right). Only signal and a small residual hyperon contribution survive the `BDTg`>0.96 cut.

B Selection Efficiency as a Function of True Momentum

The selection efficiency is evaluated on the signal Monte Carlo sample as a function of the true momentum of each final-state particle in the decay chain $\bar{\nu}_\mu + \mathrm{Ar} \to \mu^+ + \Sigma^0$, $\Sigma^0 \to \Lambda + \gamma$, $\Lambda \to p + \pi^-$. Two cumulative selection levels are shown:

- **Pre-MVA**: all pre-selection and topological distance cuts applied (cut level 10 in the sequence of Section 4).
- **Final**: additionally requiring BDTg > 0.96.

The denominator in all cases is the set of true signal events satisfying the truth-level fiducial volume requirement (Section 4). The efficiency and its 68% Bayesian (uniform prior) confidence interval are computed with `TEfficiency`.

Individual particle efficiencies

The efficiency rises steeply at low momentum, reflecting the kinematic thresholds in the signal definition ($p_p > 0.3\,\mathrm{GeV}/c$, $p_{\pi^-} > 0.1\,\mathrm{GeV}/c$) and the minimum track-length requirements for muon and pion identification. The BDTg cut reduces the pre-MVA efficiency by roughly a factor of two, while providing strong background suppression as discussed in Section 6.

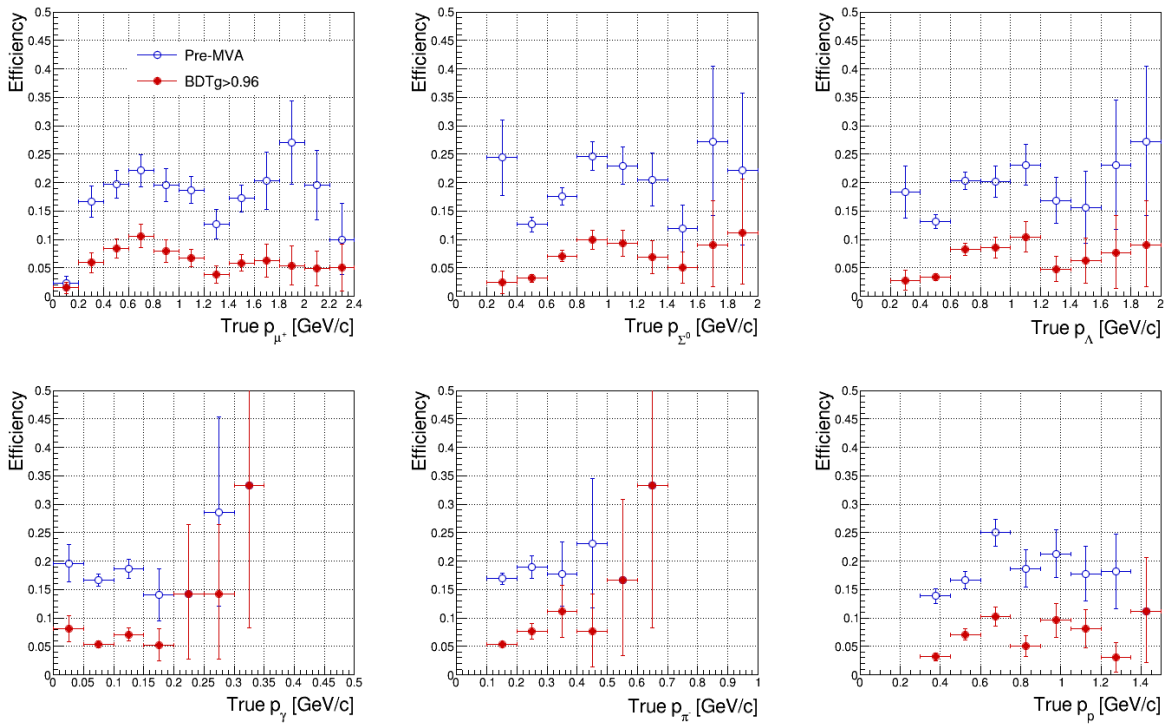


Figure 28: Selection efficiency as a function of true momentum for all six final-state particles. Blue open circles: after all pre-selection and distance cuts. Red filled circles: after the final BDTg > 0.96 requirement. Error bars indicate 68% Bayesian confidence intervals.

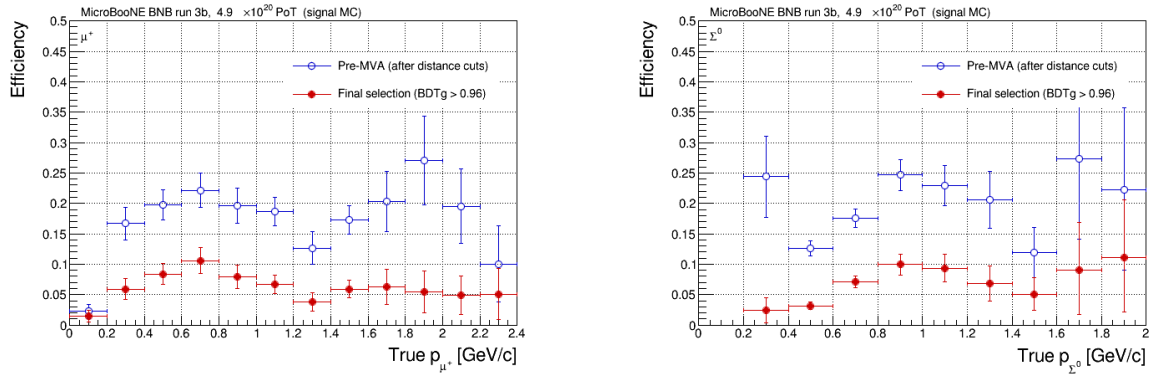


Figure 29: Left: efficiency vs. true μ^+ momentum. Right: efficiency vs. true Σ^0 momentum.

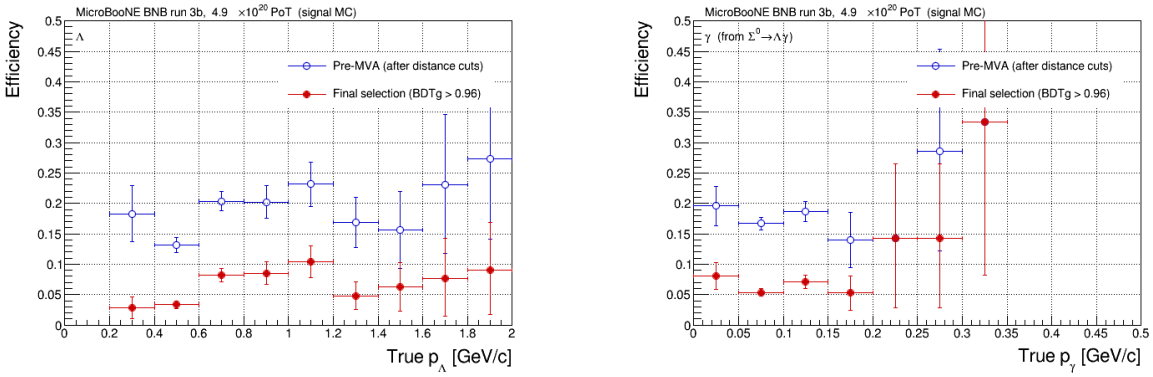


Figure 30: Left: efficiency vs. true Λ momentum. Right: efficiency vs. true photon momentum from $\Sigma^0 \to \Lambda\gamma$.

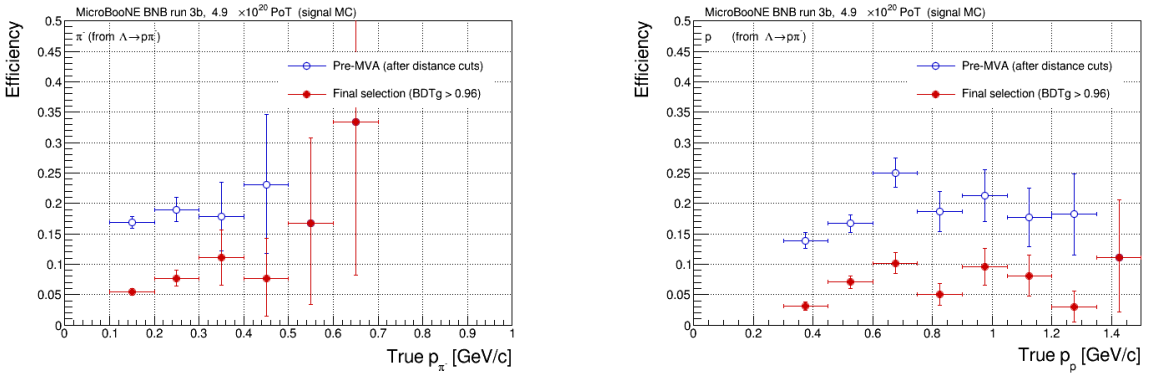


Figure 31: Left: efficiency vs. true π^- momentum from $\Lambda \to p\pi^-$. Right: efficiency vs. true proton momentum.